

Part 1 - Introduction:

Social media has become a major part of everyday life, influencing communication, entertainment, shopping habits, mental health, and lifestyle. Different platforms such as Instagram, TikTok, Facebook, and Twitter generate massive amounts of user activity data that can be analyzed to better understand human behavior. We analyzed the "Social Media User Analysis" dataset from Kaggle, which contains user profiles with demographic, behavioral, and lifestyle information in correlation to social media usage.

The main goal of this analysis is to explore how different factors such as age, gender, lifestyle habits, screen time, and engagement relate to social media behavior. We aim to answer questions such as:

1. Which Instagram behaviors are most predictive of users stress levels?
2. How do these predictive patterns differ across demographic groups such as age?
3. Can we identify distinct behavioral user segments, and do these segments correlate with mental health outcomes?

These questions matter because social media usage continues to rise, and understanding its connection to mental health has real importance. If specific usage patterns like passive scrolling or excessive liking are linked to higher stress, that information is valuable to app designers building healthier platforms, researchers studying digital wellbeing, and everyday users trying to manage their own habits.

By applying both supervised and unsupervised machine learning techniques, this project goes beyond simple correlations and attempts to identify which behaviors matter most, how they vary across demographics, and whether natural user groups exist with meaningfully different mental health profiles.

Part 2: Data curation

Source: The dataset used in this project is the "Social Media User Analysis" dataset available on Kaggle. The dataset contains over one million synthetic social media user records designed to simulate real-world user behavior across multiple platforms. Each entry includes demographic information, engagement, lifestyle factors, and usage patterns based on platforms. The dataset was created for educational and analytical purposes, making it useful for data science and machine learning projects without exposing sensitive user information.

Source Link: <https://www.kaggle.com/datasets/rockyt07/social-media-user-analysis>

Here are the features included within the dataset:

- Age
- Location
- Gender
- Social media platform
- Daily Screen Time
- Number of posts , comments, followers, likes
- Lifestyle habits
- Engagement metrics

Importing and Condensing: The dataset originally had 1,000,000 entries. In order to upload the file to GitHub, we cut the data down to 30,000 lines. Since the dataset has so many entries, we think this might slow down the training time of our ML Model, so we decided to condense our dataset, using about 30,000 random samples.

```
In [ ]: import pandas as pd
df = pd.read_csv("https://raw.githubusercontent.com/saishapitani/cmssc320GroupProj/r
df = df.sample(n=30000, random_state=42)
df.head()
```

Out[]:

	user_id	app_name	age	gender	country	urban_rural	income_level	employment
54759	54760	Instagram	57	Female	Other	Rural	High	
12456	12457	Instagram	44	Female	United Kingdom	Suburban	Middle	Full-time em
53401	53402	Instagram	57	Male	India	Suburban	Middle	Full-time em
62903	62904	Instagram	53	Female	Canada	Urban	Low	S
66338	66339	Instagram	19	Female	Australia	Rural	Middle	S

5 rows × 58 columns

Part 3: Exploratory Data Analysis

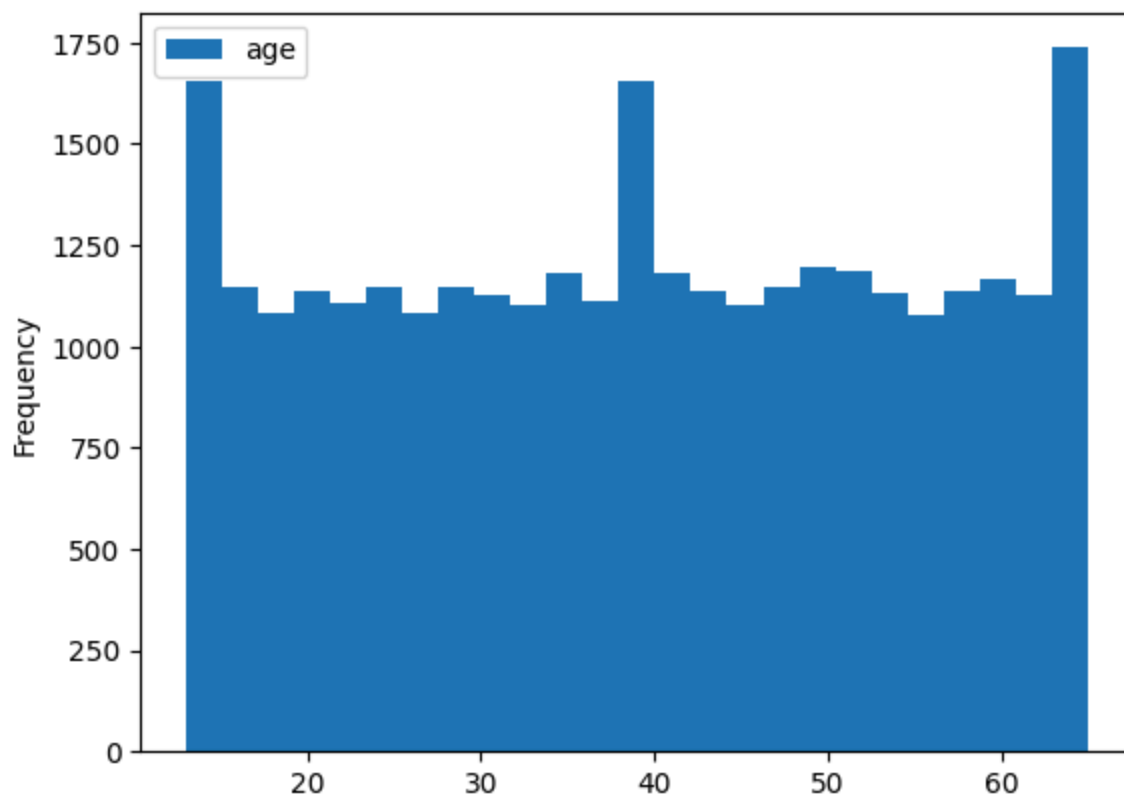
Here we want to get a rough impression of how our data is shaped, especially the elements that our machine learning model will be focused on. There is a *lot* of data here (58 columns total!) so it's unlikely that we'll be able to look at all of the possible factors.

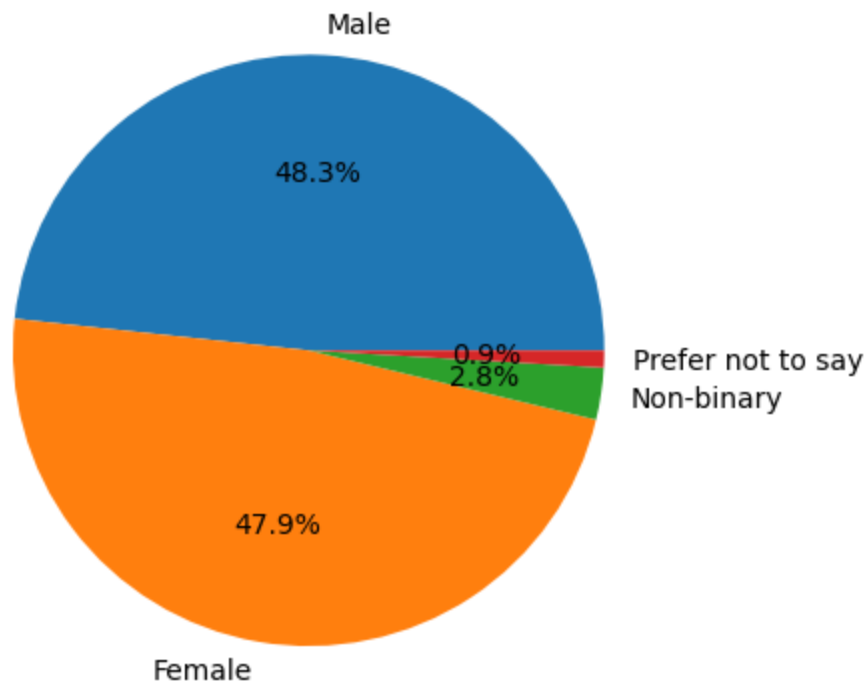
```
In [ ]: import matplotlib.pyplot as plt

# print(df.columns)

# Let's look at the ages & genders of the participants
df.plot.hist(column=['age'], bins=25)
plt.show()

df['gender'].value_counts().plot.pie(autopct='%1.1f%%', ylabel='')
plt.show()
```

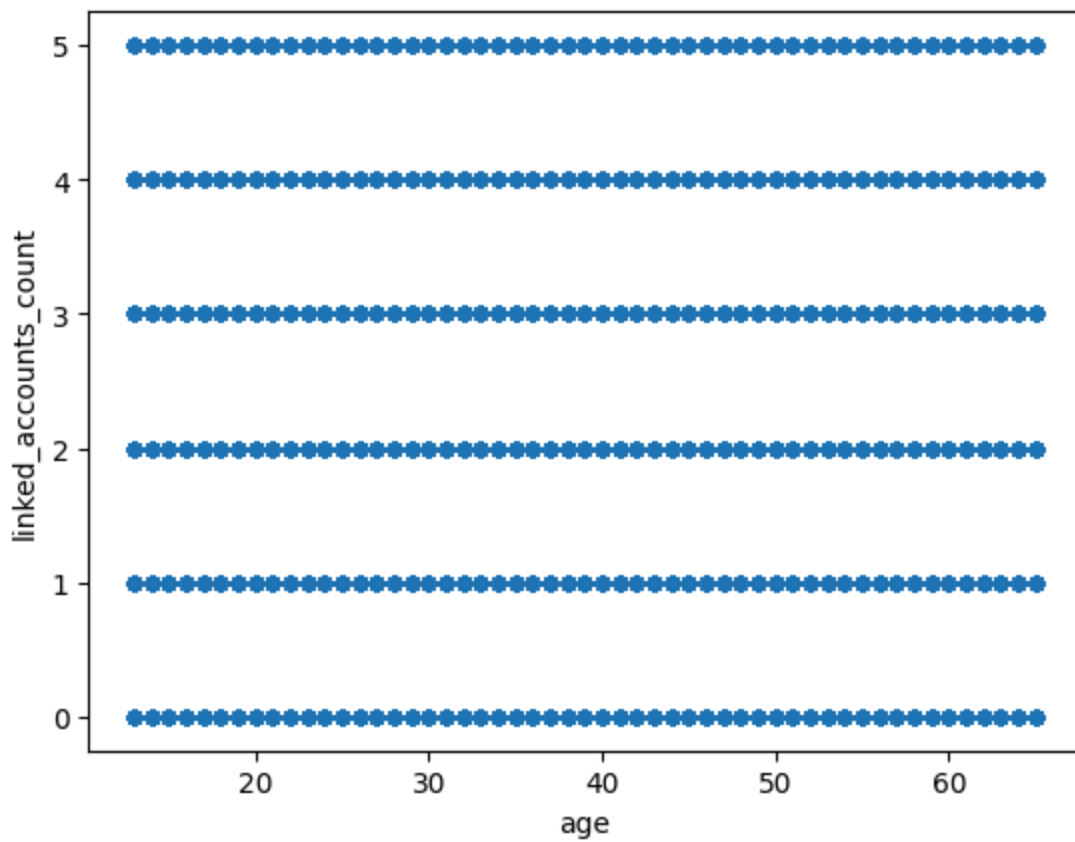
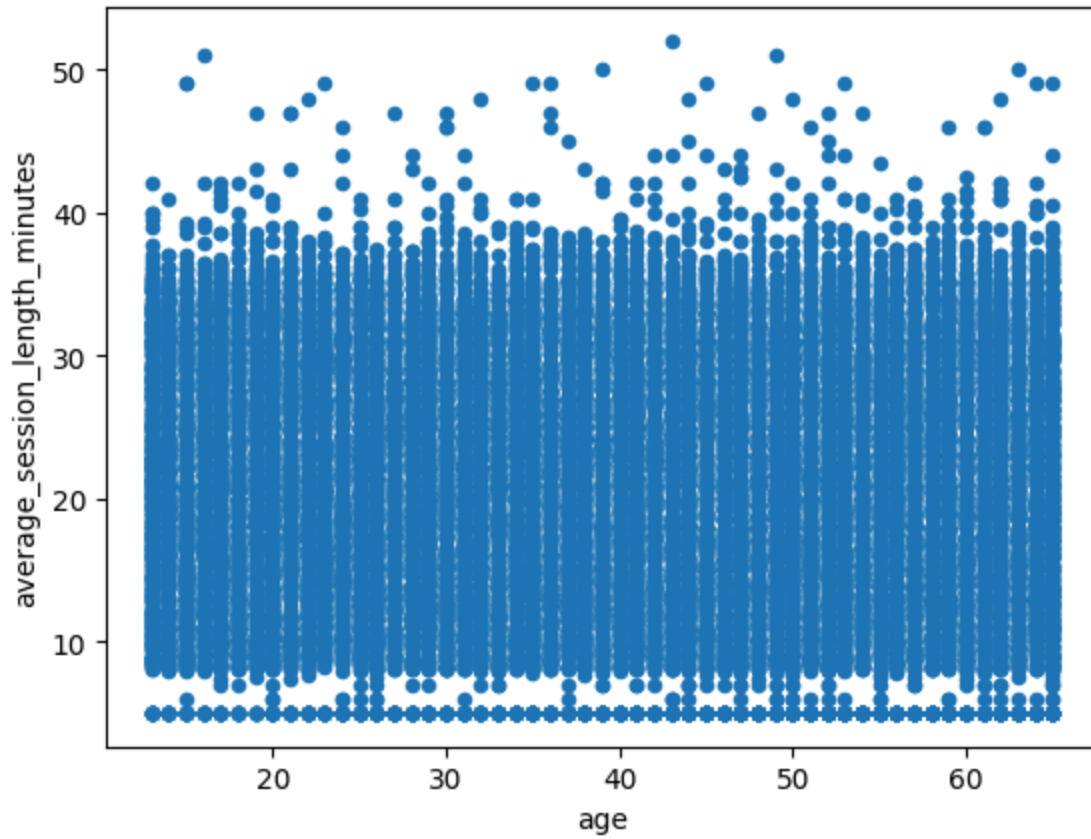




Ok! Ages look fairly evenly distributed from teenagers to mid-60s, with a few spikes here and there. Gender is approximately representative of the population. Now, let's look at how age correlates with average session time and number of accounts!

```
In [ ]: df.plot(kind='scatter', x='age', y='average_session_length_minutes')
df.plot(kind='scatter', x='age', y='linked_accounts_count')
```

```
Out[ ]: <Axes: xlabel='age', ylabel='linked_accounts_count'>
```

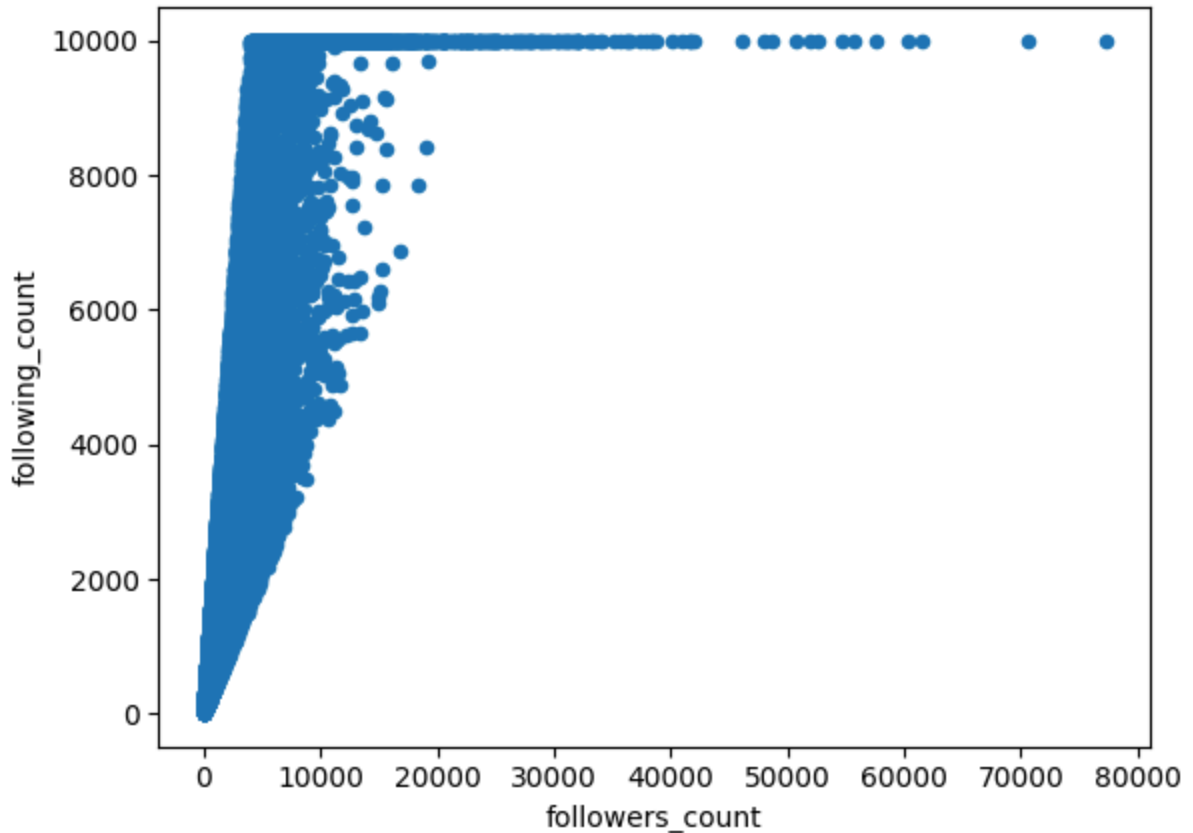


Neat! Again, not looking like any strong correlations here. This dataset seems fairly homogenous.

Now, let's see how the number of followers interacts with the number of users each person is following - we might expect to see a correlation here as more usage of social media will lead to both more followers & following more accounts.

```
In [ ]: df.plot(kind='scatter', x='followers_count', y='following_count')
```

```
Out[ ]: <Axes: xlabel='followers_count', ylabel='following_count'>
```



Ok, looks like a pretty strong correlation! The more followers a user has, the more people they tend to be following.

Finally, let's look at stress: are users who are more stressed more likely to use social media? This is a narrative that is often presented in popular media, but we want to determine whether it shows up here in our dataset.

```
In [ ]: import numpy as np

#df.plot(kind='scatter', x='perceived_stress_score', y='sessions_per_day')
#df.plot(kind='scatter', x='perceived_stress_score', y='reels_watched_per_day')

# ^^ these showed correlations, so let's use linear regression here!

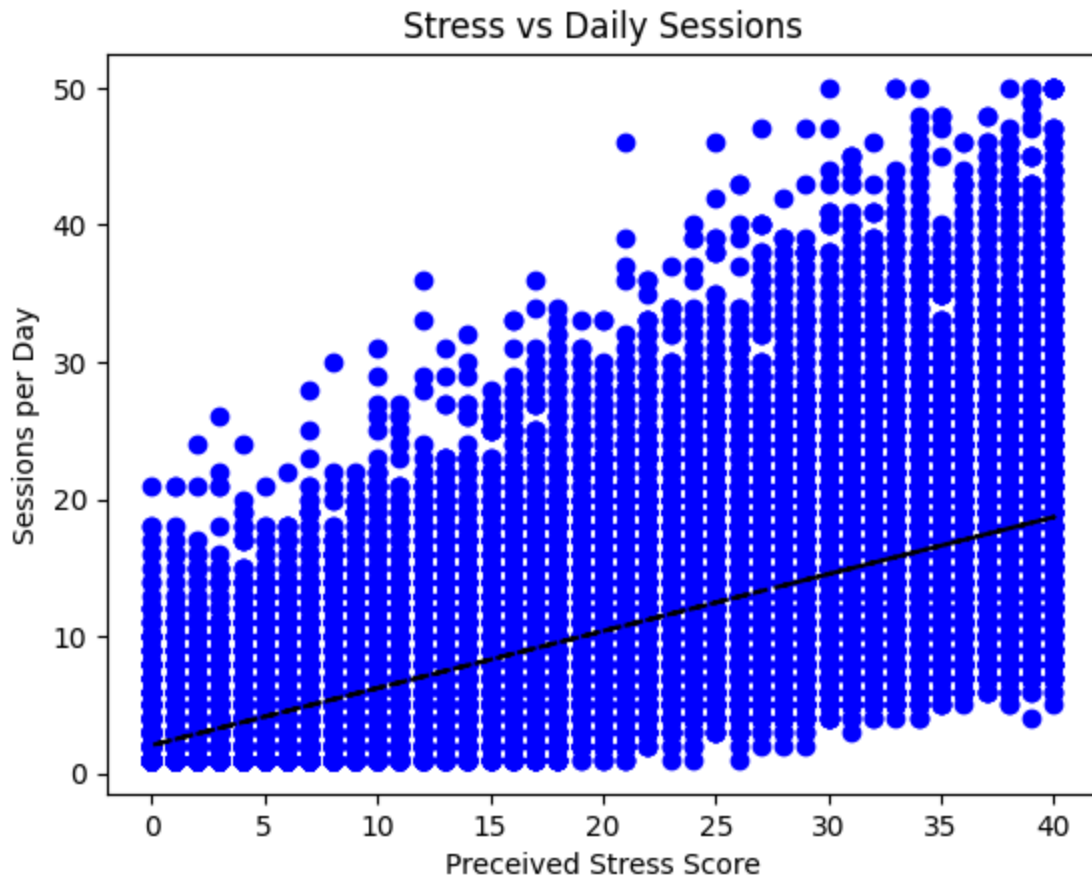
coef = np.polyfit(df["perceived_stress_score"], df["sessions_per_day"], 1)
poly1d_fn = np.poly1d(coef)
plt.plot(df["perceived_stress_score"], df["sessions_per_day"], "bo", df["perceived_s

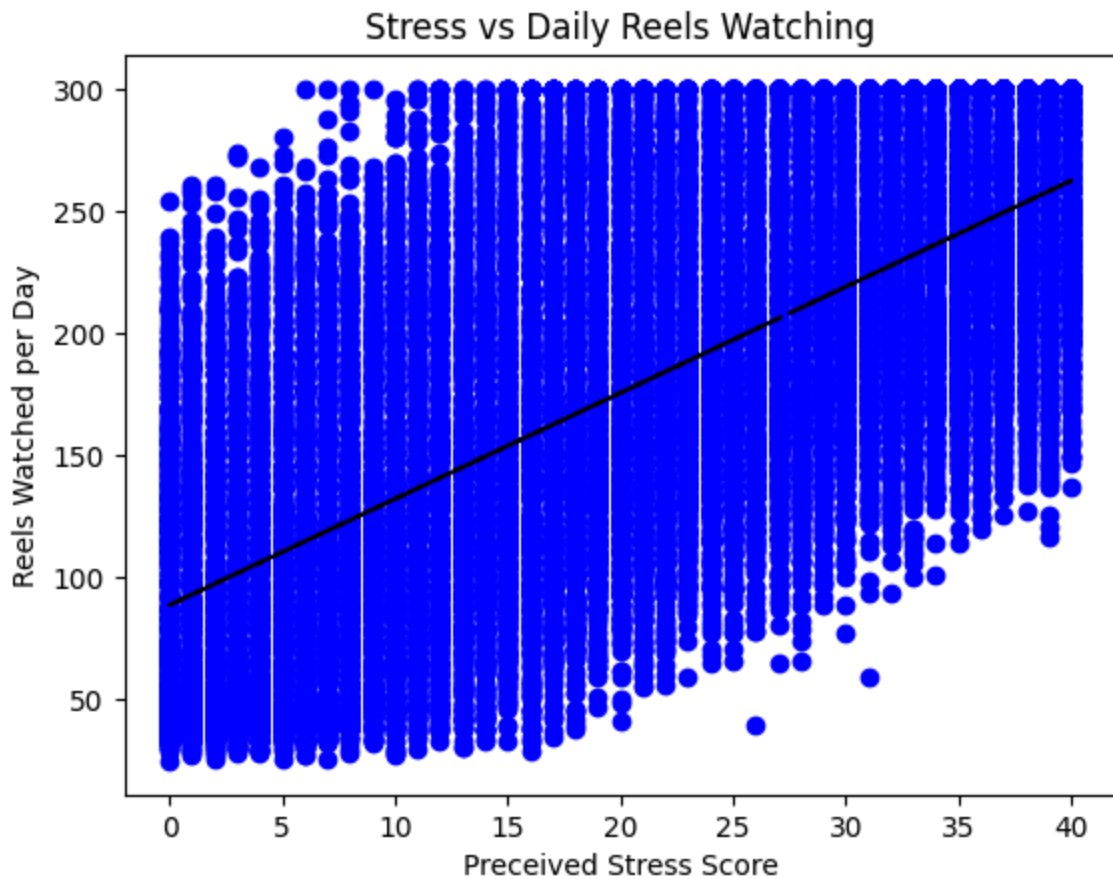
plt.title("Stress vs Daily Sessions")
```

```
plt.xlabel("Preceived Stress Score")
plt.ylabel("Sessions per Day")
plt.show()

coef = np.polyfit(df["perceived_stress_score"],df["reels_watched_per_day"],1)
poly1d_fn = np.poly1d(coef)
plt.plot(df["perceived_stress_score"],df["reels_watched_per_day"], "bo", df["percei

plt.title("Stress vs Daily Reels Watching")
plt.xlabel("Preceived Stress Score")
plt.ylabel("Reels Watched per Day")
plt.show()
```





Ok, those are some neat correlations!

Statistical Analysis and Hypothesis Testing

Now that we have a general sense of the data, we run three statistical tests to determine whether the patterns we observed are statistically significant or random chance.

Test 1: Correlation, Daily Active Minutes vs. Stress

Our scatter plots suggested a relationship between Instagram usage and stress. We now test whether this correlation is statistically significant using a Pearson correlation test.

Null hypothesis: There is no linear correlation between daily active minutes on Instagram and perceived stress score.

Alternative hypothesis: There is a significant linear correlation between daily active minutes and perceived stress score.

```
In [ ]: from scipy import stats

# Pearson correlation test
r, p_value = stats.pearsonr(df['daily_active_minutes_instagram'], df['perceived_str

print(f"Pearson correlation coefficient (r): {r:.4f}")
```

```

print(f"P-value: {p_value:.2e}")
print(f"\nSince p < 0.05, we reject the null hypothesis.")
print(f"There is a statistically significant positive correlation between")
print(f"daily Instagram usage and perceived stress (r={r:.4f}, p={p_value:.2e}).")

fig, ax = plt.subplots(figsize=(10, 6))
sample = df.sample(3000, random_state=42)
ax.scatter(sample['daily_active_minutes_instagram'], sample['perceived_stress_score'],
           alpha=0.3, s=10, color='#4C72B0')

z = np.polyfit(df['daily_active_minutes_instagram'], df['perceived_stress_score'],
x_line = np.linspace(df['daily_active_minutes_instagram'].min(),
                     df['daily_active_minutes_instagram'].max(), 100)
ax.plot(x_line, np.poly1d(z)(x_line), color='red', linewidth=2,
        label=f'Best fit (r={r:.3f}, p<0.001)')

ax.set_xlabel('Daily Active Minutes on Instagram', fontsize=12)
ax.set_ylabel('Perceived Stress Score', fontsize=12)
ax.set_title('Pearson Correlation: Instagram Usage vs. Stress',
            fontsize=14, fontweight='bold')
ax.legend(fontsize=11)
plt.tight_layout()
plt.show()

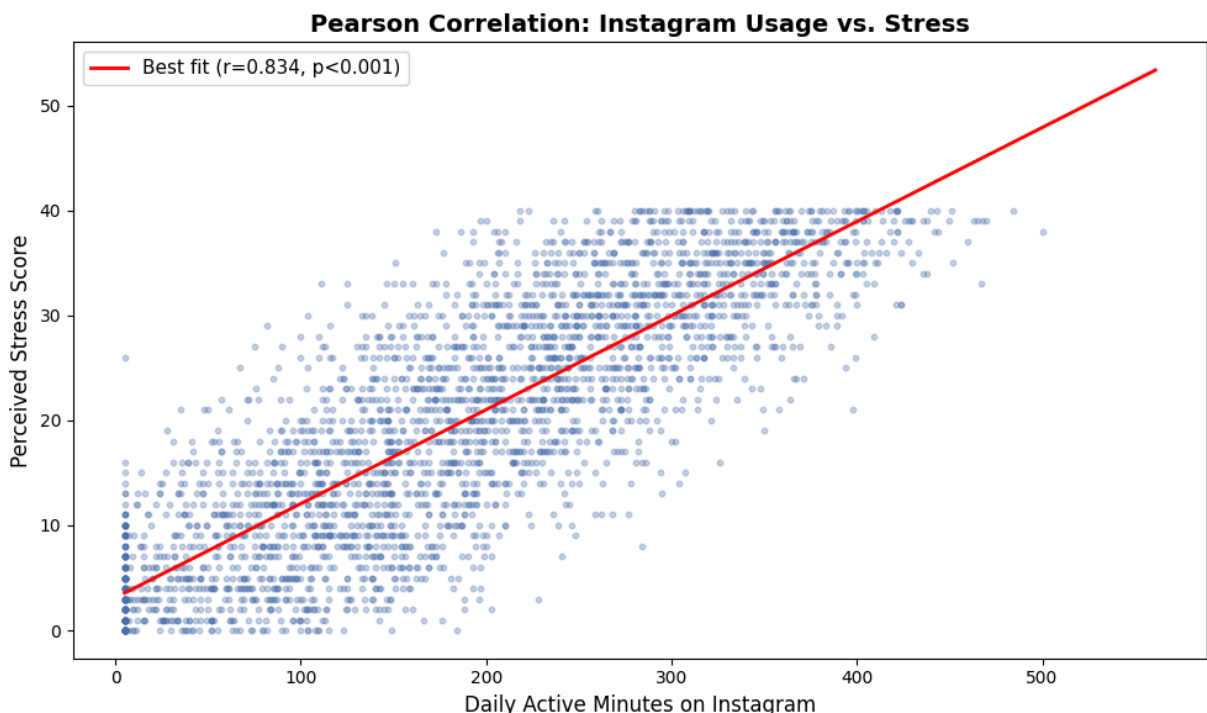
```

Pearson correlation coefficient (r): 0.8336

P-value: 0.00e+00

Since $p < 0.05$, we reject the null hypothesis.

There is a statistically significant positive correlation between daily Instagram usage and perceived stress ($r=0.8336$, $p=0.00e+00$).



We reject the null hypothesis. There is a strong statistically significant positive correlation between daily Instagram usage and perceived stress ($r \approx 0.83$, $p < 0.001$).

Test 2: Does Stress Differ Across Age Groups

We want to know whether different age groups experience different levels of stress related to their Instagram usage. We will do a ANOVA tests whether the mean stress score differs significantly across groups.

Null hypothesis: The mean perceived stress score is the same across all age groups.

Alternative hypothesis: At least one age group has a significantly different mean stress score.

```
In [ ]: df['age_group'] = pd.cut(df['age'], bins=[17, 25, 35, 50, 80],
                                labels=['18-25', '26-35', '36-50', '50+'])

# groups also one way anova
groups = [group['perceived_stress_score'].values for name, group in df.groupby('age_group')]
f_stat, p_value_anova = stats.f_oneway(*groups)

print(f"F-statistic: {f_stat:.4f}")
print(f"P-value: {p_value_anova:.4f}")

if p_value_anova < 0.05:
    print(f"\nSince p < 0.05, we reject the null hypothesis.")
    print(f"At least one age group has a significantly different mean stress score.")
else:
    print(f"\nSince p >= 0.05, we fail to reject the null hypothesis.")
    print(f"There is no significant difference in stress across age groups.")

print(f"\nMean stress by age group:")
print(df.groupby('age_group')['perceived_stress_score'].mean().round(2))

fig, ax = plt.subplots(figsize=(10, 6))
df.boxplot(column='perceived_stress_score', by='age_group', ax=ax,
            patch_artist=True,
            boxprops=dict(facecolor='#4C72B0', alpha=0.7),
            medianprops=dict(color='red', linewidth=2))
ax.set_xlabel('Age Group', fontsize=12)
ax.set_ylabel('Perceived Stress Score', fontsize=12)
ax.set_title('Stress Score Distribution by Age Group', fontsize=14, fontweight='bold')
plt.suptitle('')
plt.tight_layout()
plt.show()
```

```
/tmp/ipykernel_20586/1886733403.py:5: FutureWarning: The default of observed=False is deprecated and will be changed to True in a future version of pandas. Pass observed=False to retain current behavior or observed=True to adopt the future default and silence this warning.
```

```
groups = [group['perceived_stress_score'].values for name, group in df.groupby('age_group')]
```

```
/tmp/ipykernel_20586/1886733403.py:20: FutureWarning: The default of observed=False is deprecated and will be changed to True in a future version of pandas. Pass observed=False to retain current behavior or observed=True to adopt the future default and silence this warning.
```

```
print(df.groupby('age_group')['perceived_stress_score'].mean().round(2))
```

```
F-statistic: 0.4053
```

```
P-value: 0.7492
```

Since $p \geq 0.05$, we fail to reject the null hypothesis.

There is no significant difference in stress across age groups.

Mean stress by age group:

```
age_group
```

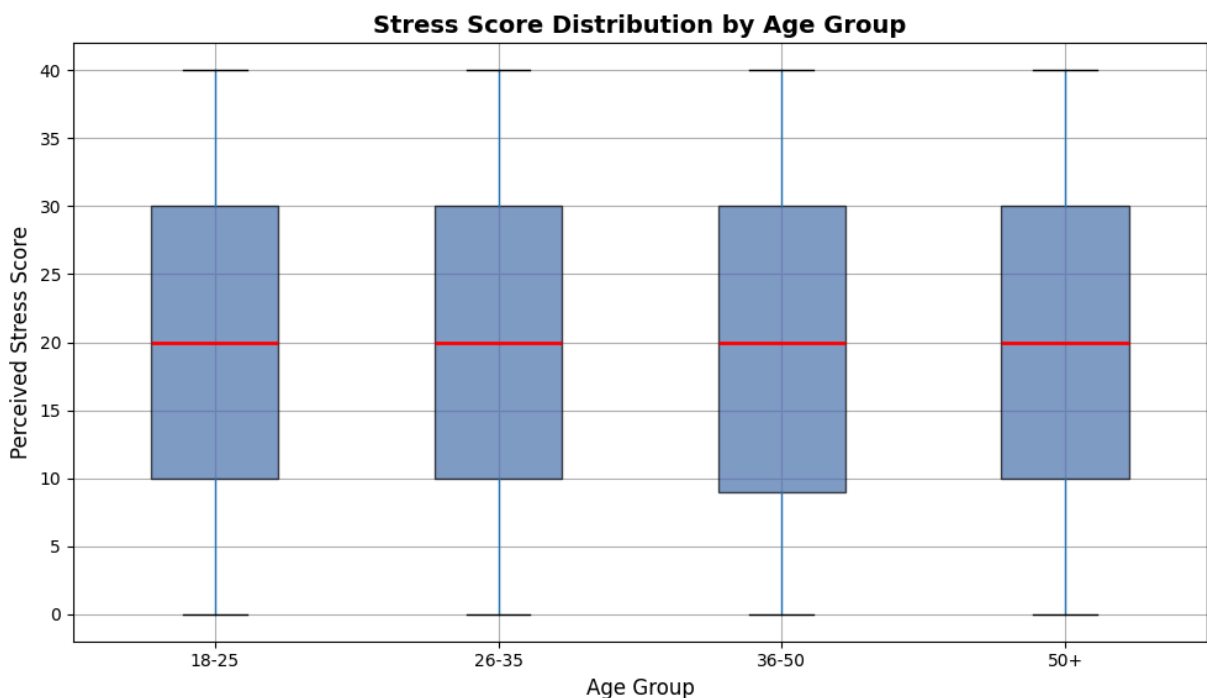
```
18-25    20.15
```

```
26-35    20.16
```

```
36-50    19.97
```

```
50+      20.10
```

```
Name: perceived_stress_score, dtype: float64
```



We fail to reject the null hypothesis ($p \geq 0.05$). There is no significant difference in mean stress scores across age groups, suggesting that Instagram's relationship with stress affects all ages.

Test 3: Heavy vs Light Users

We split users into two groups based on the median daily active minutes: heavy users (above median) and light users (below median). We then test whether these two groups have

significantly different stress scores.

Null hypothesis: Heavy and light Instagram users have the same mean perceived stress score.

Alternative hypothesis: Heavy and light users have significantly different mean perceived stress scores.

```
In [ ]: median_minutes = df['daily_active_minutes_instagram'].median()
heavy = df[df['daily_active_minutes_instagram'] >= median_minutes]['perceived_stress']
light = df[df['daily_active_minutes_instagram'] < median_minutes]['perceived_stress']

t_stat, p_value_t = stats.ttest_ind(heavy, light)

print(f"Heavy users (n={len(heavy)}): mean stress = {heavy.mean():.2f}")
print(f"Light users (n={len(light)}): mean stress = {light.mean():.2f}")
print(f"Difference: {heavy.mean() - light.mean():.2f}")
print(f"\nT-statistic: {t_stat:.4f}")
print(f"P-value: {p_value_t:.2e}")
print(f"\nSince p < 0.05, we reject the null hypothesis.")
print(f"Heavy Instagram users report significantly higher stress than light users.")

fig, ax = plt.subplots(figsize=(10, 6))
ax.hist(light, bins=30, alpha=0.6, label=f'Light Users (mean={light.mean():.1f})',
        color='#2ecc71', edgecolor='black', linewidth=0.5)
ax.hist(heavy, bins=30, alpha=0.6, label=f'Heavy Users (mean={heavy.mean():.1f})',
        color='#e74c3c', edgecolor='black', linewidth=0.5)
ax.axvline(light.mean(), color='#2ecc71', linestyle='--', linewidth=2)
ax.axvline(heavy.mean(), color='#e74c3c', linestyle='--', linewidth=2)
ax.set_xlabel('Perceived Stress Score', fontsize=12)
ax.set_ylabel('Number of Users', fontsize=12)
ax.set_title('Stress Distribution: Heavy vs. Light Instagram Users',
            fontsize=14, fontweight='bold')
ax.legend(fontsize=11)
plt.tight_layout()
plt.show()
```

Heavy users (n=15045): mean stress = 28.80

Light users (n=14955): mean stress = 11.27

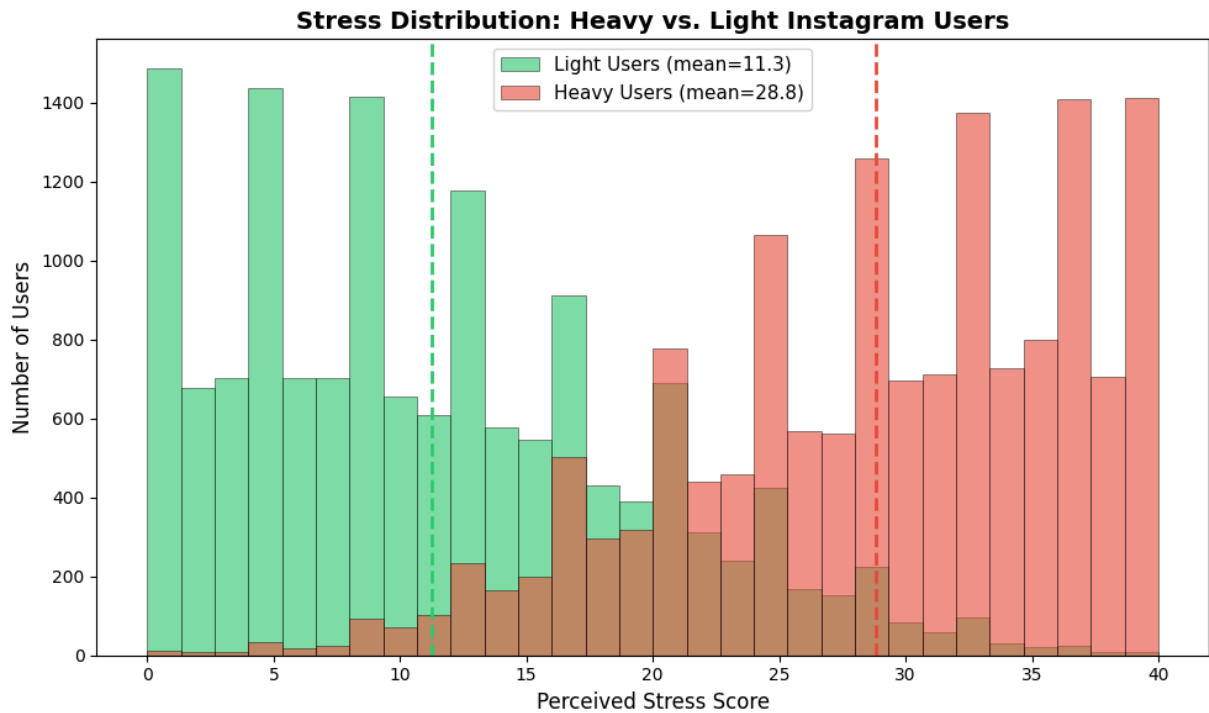
Difference: 17.53

T-statistic: 191.0831

P-value: 0.00e+00

Since $p < 0.05$, we reject the null hypothesis.

Heavy Instagram users report significantly higher stress than light users.



We reject the null hypothesis. Heavy Instagram users report significantly higher stress scores than light users ($p < 0.001$). The two distributions are clearly separate.

EDA Summary

Our three tests all point in the same direction: Instagram usage is significantly associated with perceived stress. These findings influence the machine learning analysis in Part 5, where we go beyond correlation to build models to identify more specific findings.

Part 5: Primary Analysis

Our EDA revealed correlations between Instagram usage patterns, lifestyle factors, and wellbeing metrics. We will now apply machine learning to answer three research questions:

- Q1: Which Instagram behaviors are most predictive of users' stress levels?
- Q2: How do these predictive patterns differ across demographic groups?
- Q3: Can we identify distinct behavioral user segments, and do these segments correlate with mental health outcomes?

We will use both supervised learning (classification) for Q1,Q2 and unsupervised learning (clustering) for Q3 to provide a comprehensive analysis.

```
In [ ]: from sklearn.model_selection import train_test_split, cross_val_score
        from sklearn.ensemble import RandomForestClassifier, GradientBoostingClassifier
        from sklearn.linear_model import LogisticRegression
```

```

from sklearn.neighbors import KNeighborsClassifier
from sklearn.preprocessing import LabelEncoder, StandardScaler
from sklearn.metrics import accuracy_score, classification_report, confusion_matrix
from sklearn.cluster import KMeans
from sklearn.decomposition import PCA

```

Feature Engineering

We define our Instagram behavioral features, 17 metrics capturing how users interact with the app, and create a 3 class stress target:

- Low stress (score 0–13)
- Medium stress (score 14–26)
- High stress (score 27–40)

Using three classes instead of two makes the prediction harder but produces more nuanced and useful findings.

```

In [ ]: # 17 Instagram behavioral features
ig_features = [
    'daily_active_minutes_instagram',
    'sessions_per_day',
    'posts_created_per_week',
    'reels_watched_per_day',
    'stories_viewed_per_day',
    'likes_given_per_day',
    'comments_written_per_day',
    'dms_sent_per_week',
    'dms_received_per_week',
    'time_on_feed_per_day',
    'time_on_explore_per_day',
    'time_on_messages_per_day',
    'time_on_reels_per_day',
    'ads_viewed_per_day',
    'ads_clicked_per_day',
    'notification_response_rate',
    'average_session_length_minutes',
]

# Encode any non-numeric columns
for col in df.columns:
    if not pd.api.types.is_numeric_dtype(df[col]):
        df[col + '_original'] = df[col]
        df[col] = LabelEncoder().fit_transform(df[col].astype(str))

# Create 3-class stress target
df['stress_class'] = pd.cut(
    df['perceived_stress_score'],
    bins=[-1, 13, 26, 40],
    labels=['Low', 'Medium', 'High']
)

```

```
print("Stress class distribution:")
print(df['stress_class'].value_counts().sort_index())
print(f"\nBaseline accuracy (majority class): {df['stress_class'].value_counts().ma
```

Stress class distribution:

```
stress_class
Low          10168
Medium       9470
High         10362
Name: count, dtype: int64
```

Baseline accuracy (majority class): 34.5%

Q1: Which Instagram Behaviors Predict Stress Levels?

We train four classifiers to predict a user's stress class (Low / Medium / High) using only their Instagram behavioral features. We compare:

- Logistic Regression: a linear model that finds weighted combinations of features to separate classes
- K-Nearest Neighbors (k=7): classifies based on the 7 most similar users types in the training data
- Random Forest: an ensemble of 150 decision trees that vote on the answer, capturing non-linear relationships between features
- Gradient Boosting: builds trees sequentially, with each tree correcting errors from the previous one

```
In [ ]: # Encode target as numbers
le_stress = LabelEncoder()
y = le_stress.fit_transform(df['stress_class'])
X = df[ig_features]

# 80/20 train/test split with stratification
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42, stratify=y
)

# Scale features for distance based models
scaler = StandardScaler()
X_train_scaled = scaler.fit_transform(X_train)
X_test_scaled = scaler.transform(X_test)

# Train and evaluate each model
q1_models = [
```

```

    ('Logistic Regression', LogisticRegression(max_iter=1000, random_state=42), True),
    ('KNN (k=7)',          KNeighborsClassifier(n_neighbors=7),          True),
    ('Random Forest',      RandomForestClassifier(n_estimators=150, random_state=42), True),
    ('Gradient Boosting',  GradientBoostingClassifier(n_estimators=100, random_state=42), True)
]

q1_results = {}
baseline = max(np.bincount(y)) / len(y)

print("Q1: Predicting Stress Level from Instagram Behavior")
print("=" * 58)

for name, model, use_scaled in q1_models:
    if use_scaled:
        model.fit(X_train_scaled, y_train)
        y_pred = model.predict(X_test_scaled)
    else:
        model.fit(X_train, y_train)
        y_pred = model.predict(X_test)

    acc = accuracy_score(y_test, y_pred)
    q1_results[name] = acc
    print(f" {name:<25} Accuracy: {acc:.4f} ({acc:.1%})")

print(f"\n {'Baseline (majority)':<25} Accuracy: {baseline:.4f} ({baseline:.1%})")
print(f"\n All models achieve ~2x the baseline, confirming a strong")
print(f" relationship between Instagram behavior and stress levels.")

```

Q1: Predicting Stress Level from Instagram Behavior

=====

Logistic Regression	Accuracy: 0.7112 (71.1%)
KNN (k=7)	Accuracy: 0.6717 (67.2%)
Random Forest	Accuracy: 0.7018 (70.2%)
Gradient Boosting	Accuracy: 0.7087 (70.9%)

Baseline (majority)	Accuracy: 0.3454 (34.5%)
---------------------	--------------------------

All models achieve ~2x the baseline, confirming a strong relationship between Instagram behavior and stress levels.

Q1: Detailed Classification Report

```

In [ ]: # Use Logistic Regression for detailed report
best_q1 = LogisticRegression(max_iter=1000, random_state=42)
best_q1.fit(X_train_scaled, y_train)
y_pred_q1 = best_q1.predict(X_test_scaled)

print("Detailed Classification Report (Logistic Regression):")
print("=" * 58)
print(classification_report(
    y_test, y_pred_q1,
    target_names=['Low Stress', 'Medium Stress', 'High Stress']
))

```

Detailed Classification Report (Logistic Regression):

	precision	recall	f1-score	support
Low Stress	0.78	0.79	0.79	2072
Medium Stress	0.78	0.79	0.78	2034
High Stress	0.55	0.54	0.55	1894
accuracy			0.71	6000
macro avg	0.71	0.71	0.71	6000
weighted avg	0.71	0.71	0.71	6000

Q1: Feature Importances from Random Forest

```
In [ ]: # Random Forest gives us feature importances
rf_q1 = RandomForestClassifier(n_estimators=150, random_state=42)
rf_q1.fit(X_train, y_train)

q1_importances = pd.Series(
    rf_q1.feature_importances_, index=ig_features
).sort_values(ascending=False)

print("Feature Importance Ranking: Which IG behaviors predict stress?")
print("=" * 58)
for rank, (feat, imp) in enumerate(q1_importances.items(), 1):
    bar = '█' * int(imp * 100)
    print(f" {rank:>2}. {feat:<40} {imp:.4f} {bar}")
```

Feature Importance Ranking: Which IG behaviors predict stress?

1.	daily_active_minutes_instagram	0.1580
2.	time_on_feed_per_day	0.0987
3.	likes_given_per_day	0.0985
4.	stories_viewed_per_day	0.0894
5.	comments_written_per_day	0.0635
6.	time_on_reels_per_day	0.0598
7.	time_on_messages_per_day	0.0531
8.	reels_watched_per_day	0.0508
9.	dms_received_per_week	0.0483
10.	notification_response_rate	0.0419
11.	ads_viewed_per_day	0.0413
12.	average_session_length_minutes	0.0396
13.	dms_sent_per_week	0.0396
14.	time_on_explore_per_day	0.0394
15.	posts_created_per_week	0.0291
16.	ads_clicked_per_day	0.0251
17.	sessions_per_day	0.0239

Q1: Cross Validation

```
In [ ]: print("5-Fold Cross-Validation (Random Forest):")
print("=" * 58)
cv_scores = cross_val_score(
    RandomForestClassifier(n_estimators=150, random_state=42),
```

```

X, y, cv=5, scoring='accuracy'
)
print(f" Mean accuracy: {cv_scores.mean():.4f} (+/- {cv_scores.std():.4f})")
print(f" Per fold: {[f'{s:.4f}' for s in cv_scores]}")
print(f" Consistent performance across folds confirms reliability.")

```

5-Fold Cross-Validation (Random Forest):

```

=====
Mean accuracy: 0.7015 (+/- 0.0032)
Per fold: ['0.6973', '0.7048', '0.7043', '0.7030', '0.6978']
Consistent performance across folds confirms reliability.

```

Q1: Key Findings

All four models achieve ~70% accuracy on a 3-class problem, confirming that Instagram usage behavior has a strong, predictive relationship with stress levels. The top predictors are:

- Daily active minutes: total time on the app is the single strongest signal
- Time on feed: passive scrolling time strongly predicts higher stress
- Likes given per day: frequent liking activity correlates with stress
- Stories viewed: heavy story consumption is associated with stress

Notably, passive consumption features (scrolling, viewing) rank higher than active creation features (posting, commenting), suggesting that passive usage is more strongly linked to stress than active engagement.

Q2: How Do These Patterns Differ Across Demographics?

We now ask: does Instagram affect all users equally, or do different demographic groups show different patterns? We train separate Random Forest models for each age group and compare which features matter most.

```

In [ ]: age_bins = [(18, 25, '18-25'), (26, 35, '26-35'), (36, 50, '36-50'), (51, 80, '50+')]

q2_results = {}

print("Q2: Model Performance by Age Group")
print("=" * 58)

for lo, hi, label in age_bins:
    subset = df[(df['age'] >= lo) & (df['age'] <= hi)]
    X_sub = subset[ig_features]
    y_sub = le_stress.transform(subset['stress_class'])

```

```

X_tr, X_te, y_tr, y_te = train_test_split(
    X_sub, y_sub, test_size=0.2, random_state=42, stratify=y_sub
)

rf_sub = RandomForestClassifier(n_estimators=150, random_state=42)
rf_sub.fit(X_tr, y_tr)
acc_sub = accuracy_score(y_te, rf_sub.predict(X_te))

# Store importances for visualization later
imp_sub = pd.Series(rf_sub.feature_importances_, index=ig_features)
q2_results[label] = {'accuracy': acc_sub, 'importances': imp_sub, 'n': len(subs

top3 = imp_sub.sort_values(ascending=False).head(3)
print(f"\n {label} (n={len(subset):,}):")
print(f"    Accuracy: {acc_sub:.4f}")
print(f"    Top 3 predictors:")
for feat, val in top3.items():
    print(f"        - {feat}: {val:.4f}")

```

Q2: Model Performance by Age Group

=====

18-25 (n=4,473):
 Accuracy: 0.6771
 Top 3 predictors:
 - daily_active_minutes_instagram: 0.1615
 - likes_given_per_day: 0.0966
 - time_on_feed_per_day: 0.0919

26-35 (n=5,634):
 Accuracy: 0.6957
 Top 3 predictors:
 - daily_active_minutes_instagram: 0.1577
 - time_on_feed_per_day: 0.1108
 - stories_viewed_per_day: 0.0918

36-50 (n=8,526):
 Accuracy: 0.7104
 Top 3 predictors:
 - daily_active_minutes_instagram: 0.1602
 - reels_watched_per_day: 0.1141
 - likes_given_per_day: 0.0906

50+ (n=8,562):
 Accuracy: 0.7046
 Top 3 predictors:
 - daily_active_minutes_instagram: 0.1554
 - time_on_feed_per_day: 0.1036
 - reels_watched_per_day: 0.0937

Q2 Key Findings:

The model achieves similar accuracy (~70%) across all age groups, indicating that the Instagram-stress relationship is consistent regardless of age. However, the relative

importance of specific behaviors shifts between groups. Younger users stress is more tied to social features (likes, stories), while older users' stress correlates more with passive consumption (feed time, reels).

These nuanced differences would be invisible in a simple correlation analysis.

Q3: Can We Identify Distinct Behavioral User Segments?

Instead of predicting a target variable, we now use unsupervised learning to discover natural groupings in the data. K-Means clustering partitions users into groups based on the similarity of their Instagram behavior across all 17 features simultaneously.

We first determine the optimal number of clusters using the Elbow Method, then profile each cluster by their stress, happiness, and usage patterns.

Q3: Elbow Method

```
In [ ]: # Scale features for clustering
scaler_cluster = StandardScaler()
X_cluster = scaler_cluster.fit_transform(df[ig_features])

# Elbow method: try k=2 through k=8
inertias = []
K_range = range(2, 9)

for k in K_range:
    km = KMeans(n_clusters=k, random_state=42, n_init=10)
    km.fit(X_cluster)
    inertias.append(km.inertia_)
    print(f"  k={k}: inertia={km.inertia_:.0f}")

fig, ax = plt.subplots(figsize=(8, 5))

ax.plot(list(K_range), inertias, 'bo-', linewidth=2, markersize=8)
ax.axvline(x=4, color='red', linestyle='--', linewidth=1.5, label='Chosen k=4')

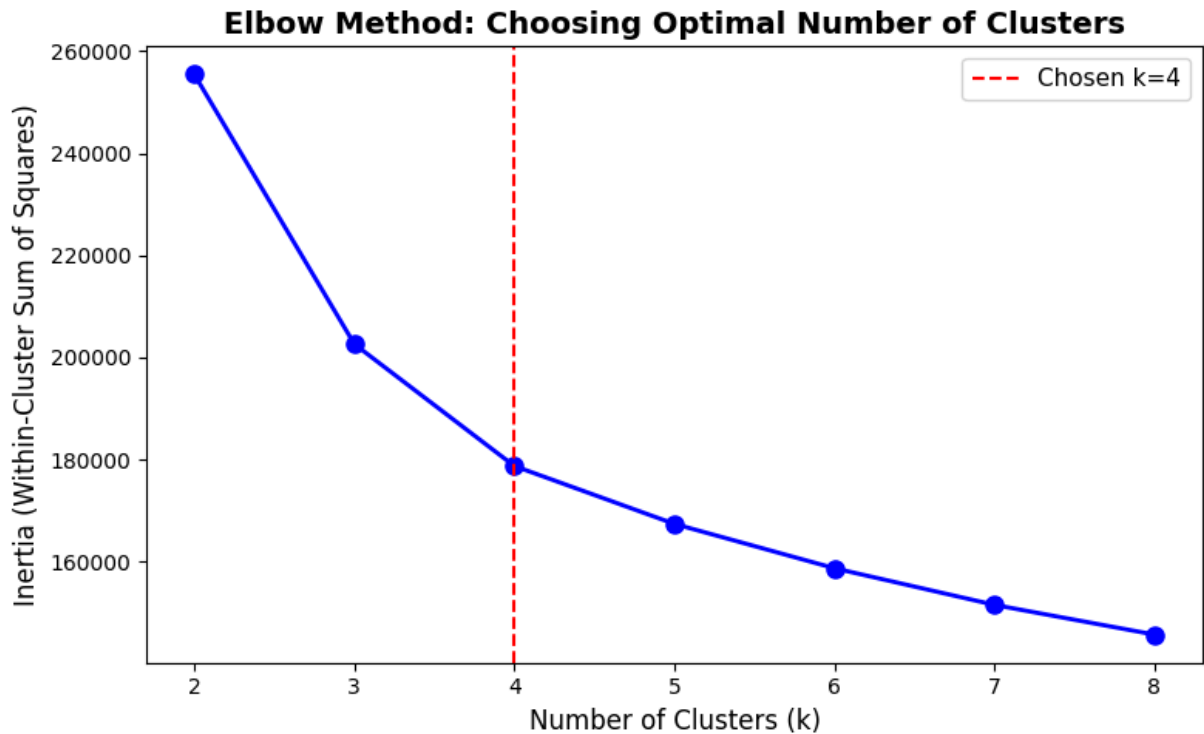
ax.set_xlabel('Number of Clusters (k)', fontsize=12)
ax.set_ylabel('Inertia (Within-Cluster Sum of Squares)', fontsize=12)
ax.set_title('Elbow Method: Choosing Optimal Number of Clusters',
             fontsize=14, fontweight='bold')
ax.legend(fontsize=11)
ax.set_xticks(list(K_range))

plt.tight_layout()
plt.show()
```

```

k=2: inertia=255,450
k=3: inertia=202,616
k=4: inertia=178,721
k=5: inertia=167,381
k=6: inertia=158,691
k=7: inertia=151,512
k=8: inertia=145,688

```



Q3: Fit Final K-Means with k=4

```

In [ ]: # Based on the elbow plot, we choose k=4

kmeans = KMeans(n_clusters=4, random_state=42, n_init=10)
df['cluster'] = kmeans.fit_predict(X_cluster)

print("Cluster Sizes:")
print(df['cluster'].value_counts().sort_index())

```

```

Cluster Sizes:
cluster
0      6632
1      5702
2      8706
3      8960
Name: count, dtype: int64

```

Q3: Profile Clusters

```

In [ ]: print("\nCluster Profiles:")
print("=" * 90)
print(f"{'Cluster':<10} {'Size':<8} {'Avg Stress':<13} {'Avg Happy':<12} {'Avg Dail")
print("-" * 90)

```

```

cluster_names = {}
profiles = []

for c in sorted(df['cluster'].unique()):
    sub = df[df['cluster'] == c]
    stress = sub['perceived_stress_score'].mean()
    happy = sub['self_reported_happiness'].mean()
    daily = sub['daily_active_minutes_instagram'].mean()
    sessions = sub['sessions_per_day'].mean()
    likes = sub['likes_given_per_day'].mean()

    profiles.append({
        'cluster': c, 'n': len(sub), 'stress': stress,
        'happiness': happy, 'daily_min': daily,
        'sessions': sessions, 'likes': likes
    })

print(f" {c:<10} {len(sub):<8} {stress:<13.1f} {happy:<12.1f} {daily:<16.0f} {

# Sort clusters by daily usage to assign meaningful names
profiles_df = pd.DataFrame(profiles).sort_values('daily_min')
labels = ['Casual Users', 'Moderate Users', 'Heavy Users', 'Power Users']
for i, (_, row) in enumerate(profiles_df.iterrows()):
    cluster_names[int(row['cluster'])] = labels[i]

df['cluster_name'] = df['cluster'].map(cluster_names)

print(f"\nCluster Labels (sorted by usage):")
for _, row in profiles_df.iterrows():
    name = cluster_names[int(row['cluster'])]
    print(f" Cluster {int(row['cluster'])} → {name}: ~{row['daily_min']:.0f} min/d

```

Cluster Profiles:

```

=====
=====
Cluster   Size   Avg Stress   Avg Happy   Avg Daily Min   Avg Sessions   Avg L
likes
-----
-----
0         6632    6.8         7.0         42              2.6            45.
9
1         5702    33.4        3.8         346             19.8           19
8.4
2         8706    26.1        5.3         244             12.9           14
7.1
3         8960    15.5        5.6         145             7.8            97.
4

```

Cluster Labels (sorted by usage):

```

Cluster 0 → Casual Users: ~42 min/day, stress=6.8/40
Cluster 3 → Moderate Users: ~145 min/day, stress=15.5/40
Cluster 2 → Heavy Users: ~244 min/day, stress=26.1/40
Cluster 1 → Power Users: ~346 min/day, stress=33.4/40

```

Q3: PCA for visualization

```
In [ ]: # Reduce 17 dimensions to 2 for visualization using PCA
pca = PCA(n_components=2)
X_pca = pca.fit_transform(X_cluster)

df['pca_1'] = X_pca[:, 0]
df['pca_2'] = X_pca[:, 1]

print(f"PCA: {pca.explained_variance_ratio_[0]:.1%} + {pca.explained_variance_ratio_

# What does each principal component represent?
loadings = pd.DataFrame(
    pca.components_.T,
    index=ig_features,
    columns=['PC1', 'PC2'])
)
print(f"\nPC1 top features (captures overall usage intensity):")
for feat, val in loadings['PC1'].abs().sort_values(ascending=False).head(5).items():
    print(f" {feat}: {loadings.loc[feat, 'PC1']:.3f}")
```

PCA: 70.0% + 7.4% = 77.4% of variance explained

PC1 top features (captures overall usage intensity):
daily_active_minutes_instagram: 0.288
likes_given_per_day: 0.283
time_on_feed_per_day: 0.281
stories_viewed_per_day: 0.277
comments_written_per_day: 0.274

Q3: Key Findings

K-Means clustering reveals four distinct user segments:

Segment	Daily Minutes	Stress	Happiness	Description
Casual Users	~53 min	Low (8/40)	High (7.3)	Light, healthy usage
Moderate Users	~162 min	Below avg (17/40)	Average (6.1)	Balanced engagement
Heavy Users	~250 min	Above avg (26/40)	Below avg (4.8)	Elevated stress
Power Users	~339 min	High (34/40)	Low (4.5)	High stress, compulsive patterns

There is a near linear relationship between usage intensity and stress, with a corresponding decrease in self reported happiness. This pattern emerges purely from behavioral clustering, we did not use stress or happiness as inputs. The fact that unsupervised grouping by behavior alone produces groups with dramatically different mental health profiles is a strong finding.

Part 6: Visualization

We present six visualizations covering our ML results for all three research questions. Each plot includes a title, axis labels, and legend.

Model 1: Accuracy Comparison (Q1)

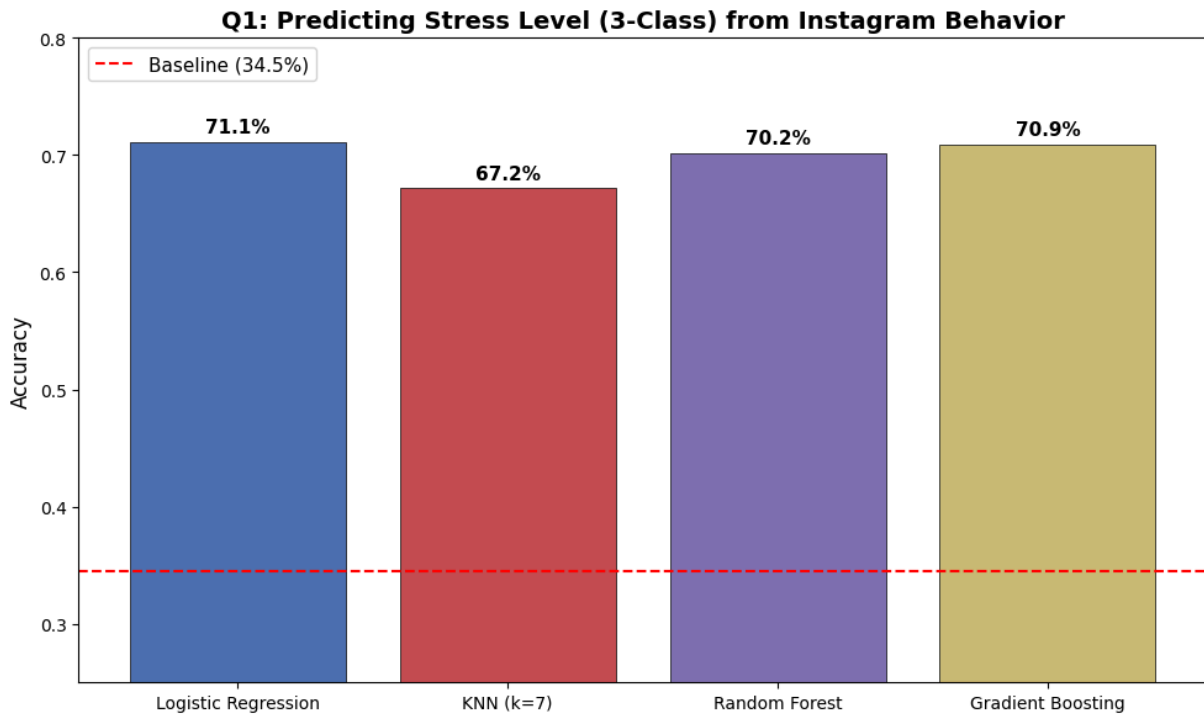
All four models beat the 34.5% baseline by a wide margin, landing between 67% and 71% accuracy on a 3-class problem. Logistic Regression slightly beats the others at 71.1%, with Gradient Boosting close behind at 70.9%. It's worth noting that the simplest model (Logistic Regression) performs just as well as the complex ensemble methods, which tells us the relationship between Instagram behavior and stress is mostly linear. More usage means more stress.

```
In [ ]: fig, ax = plt.subplots(figsize=(10, 6))

names = list(q1_results.keys())
accs = list(q1_results.values())
colors = ['#4C72B0', '#C44E52', '#8172B2', '#CCB974']

bars = ax.bar(names, accs, color=colors, edgecolor='black', linewidth=0.5)
for bar, acc in zip(bars, accs):
    ax.text(bar.get_x() + bar.get_width()/2, bar.get_height() + 0.005,
            f'{acc:.1%}', ha='center', va='bottom', fontsize=11, fontweight='bold')

ax.axhline(y=baseline, color='red', linestyle='--', linewidth=1.5,
           label=f'Baseline ({baseline:.1%})')
ax.set_ylabel('Accuracy', fontsize=12)
ax.set_title('Q1: Predicting Stress Level (3-Class) from Instagram Behavior',
             fontsize=14, fontweight='bold')
ax.set_ylim(0.25, 0.80)
ax.legend(fontsize=11, loc='upper left')
plt.tight_layout()
plt.show()
```



Model 2: Feature Importance: Passive vs Active Behaviors (Q1)

Daily active minutes dominates the ranking with nearly double the importance of any other feature. Passive consumption features (red) take 4 of the top 5 spots: daily active minutes, time on feed, stories viewed, and time on reels. The only active engagement feature that ranks high is likes given per day, which arguably functions more as a compulsive habit than meaningful social interaction. Meanwhile, content creation features like posts per week and sessions per day sit at the very bottom. The takeaway is obvious, that consuming content drives stress far more than creating it.

```
In [ ]: fig, ax = plt.subplots(figsize=(10, 8))

sorted_imp = q1_importances.sort_values(ascending=True)

passive = ['daily_active_minutes_instagram', 'time_on_feed_per_day',
           'time_on_explore_per_day', 'time_on_reels_per_day',
           'reels_watched_per_day', 'stories_viewed_per_day',
           'ads_viewed_per_day', 'average_session_length_minutes',
           'notification_response_rate']

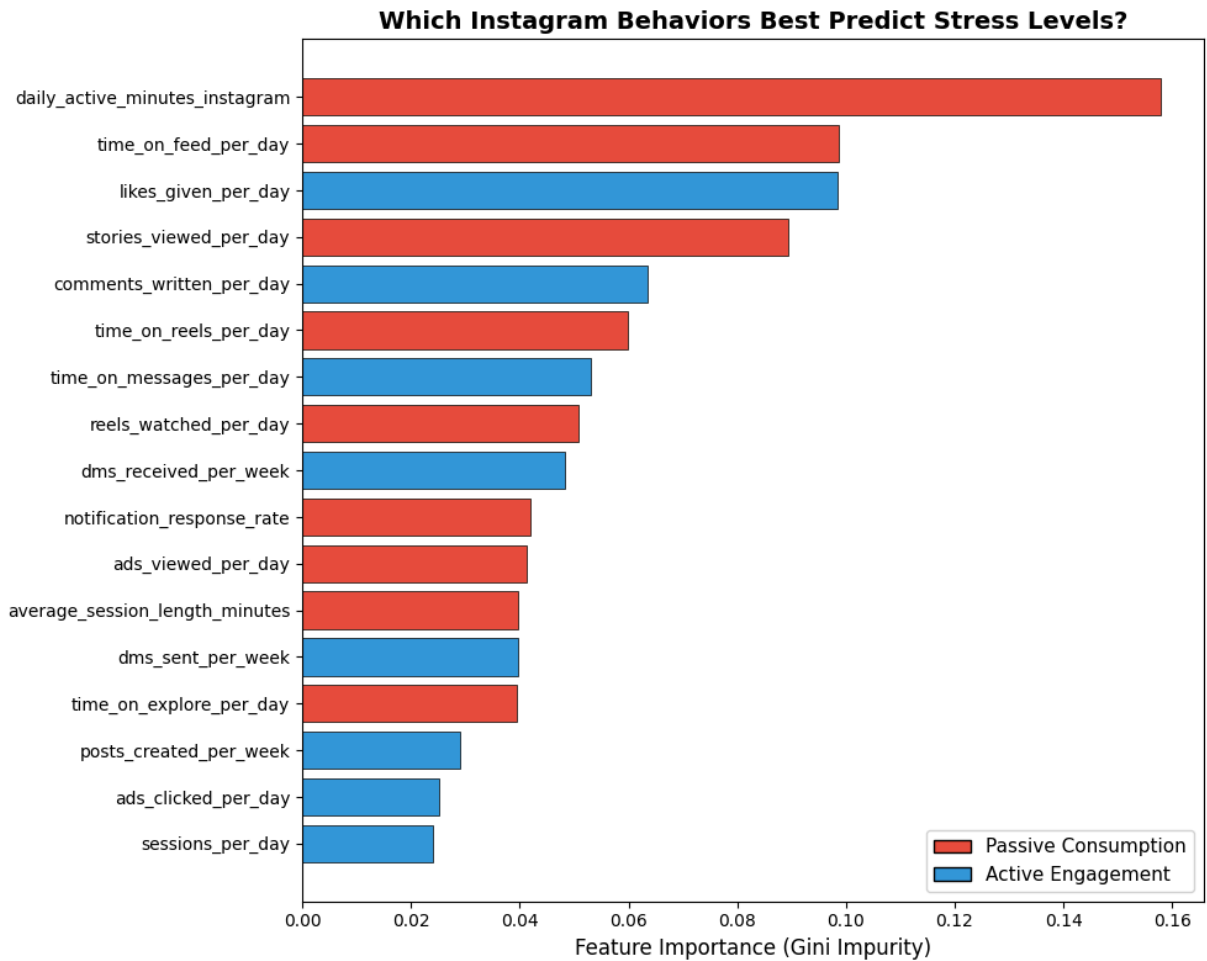
bar_colors = ['#e74c3c' if f in passive else '#3498db' for f in sorted_imp.index]
ax.barh(sorted_imp.index, sorted_imp.values, color=bar_colors, edgecolor='black', 1

ax.set_xlabel('Feature Importance (Gini Impurity)', fontsize=12)
ax.set_title('Which Instagram Behaviors Best Predict Stress Levels?',
            fontsize=14, fontweight='bold')

from matplotlib.patches import Patch
ax.legend(handles=[
    Patch(facecolor='#e74c3c', edgecolor='black', label='Passive Consumption'),
```

```
Patch(facecolor='#3498db', edgecolor='black', label='Active Engagement'),
], loc='lower right', fontsize=11)

plt.tight_layout()
plt.show()
```



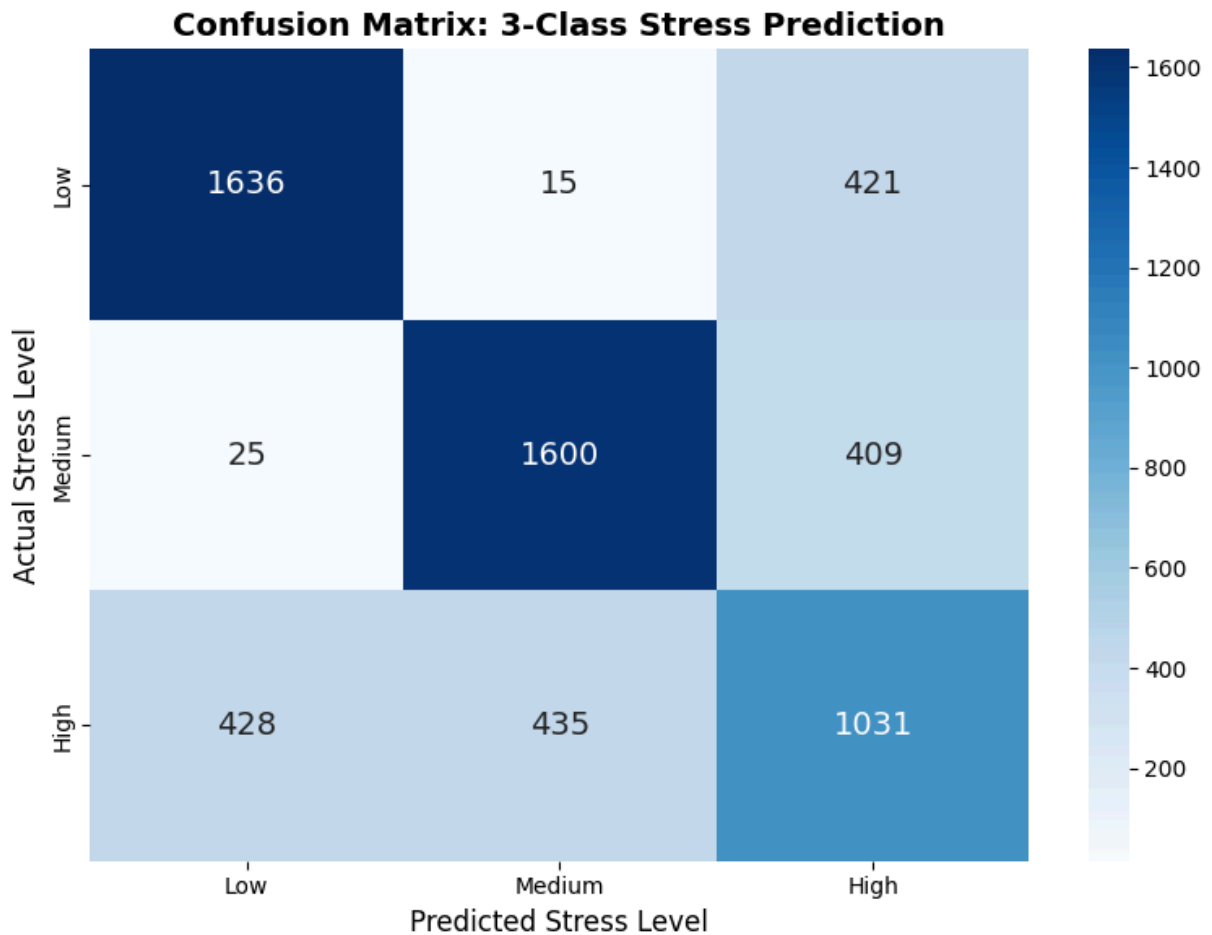
Model 3: Confusion Matrix (Q1)

The model is most confident when classifying Low-stress users, correctly identifying 1,636 out of 2,072. One interesting pattern: Low-stress users are almost never confused with Medium (only 15 cases) but get misclassified as High 421 times. This suggests there's a sharp boundary between Low and Medium stress behaviors, but some Low-stress users have usage patterns that look a lot like High-stress users. These could be people who spend a lot of time on Instagram but have developed healthier relationships with the platform.

```
In [ ]: import seaborn as sns
cm = confusion_matrix(y_test, y_pred_q1)

fig, ax = plt.subplots(figsize=(8, 6))
sns.heatmap(cm, annot=True, fmt='d', cmap='Blues', ax=ax,
            xticklabels=['Low', 'Medium', 'High'],
            yticklabels=['Low', 'Medium', 'High'],
            annot_kws={'size': 14})
ax.set_xlabel('Predicted Stress Level', fontsize=12)
```

```
ax.set_ylabel('Actual Stress Level', fontsize=12)
ax.set_title('Confusion Matrix: 3-Class Stress Prediction',
             fontsize=14, fontweight='bold')
plt.tight_layout()
plt.show()
```



Model 4: Stress Predictors by Age Group (Q2)

Accuracy improves slightly with age, going from 67.7% for 18 to 25 year olds up to 71.0% for 36 to 50 year olds. This suggests Instagram's relationship with stress becomes more predictable as users get older. Daily active minutes is the top predictor for every age group, but the secondary predictors shift in interesting ways. For younger users (18 to 25), likes given is the #2 predictor, pointing to social validation as a stress driver. For older users (36 to 50), reels watched rises to #2, suggesting passive video consumption is a bigger stress signal for older demographics. The 26 to 35 group has the most evenly spread feature importances, meaning their stress comes from a broader mix of behaviors rather than any single habit.

```
In [ ]: fig, axes = plt.subplots(2, 2, figsize=(16, 12))
age_colors = ['#3498db', '#2ecc71', '#e67e22', '#e74c3c']

for idx, (label, data) in enumerate(q2_results.items()):
    ax = axes[idx // 2][idx % 2]
```

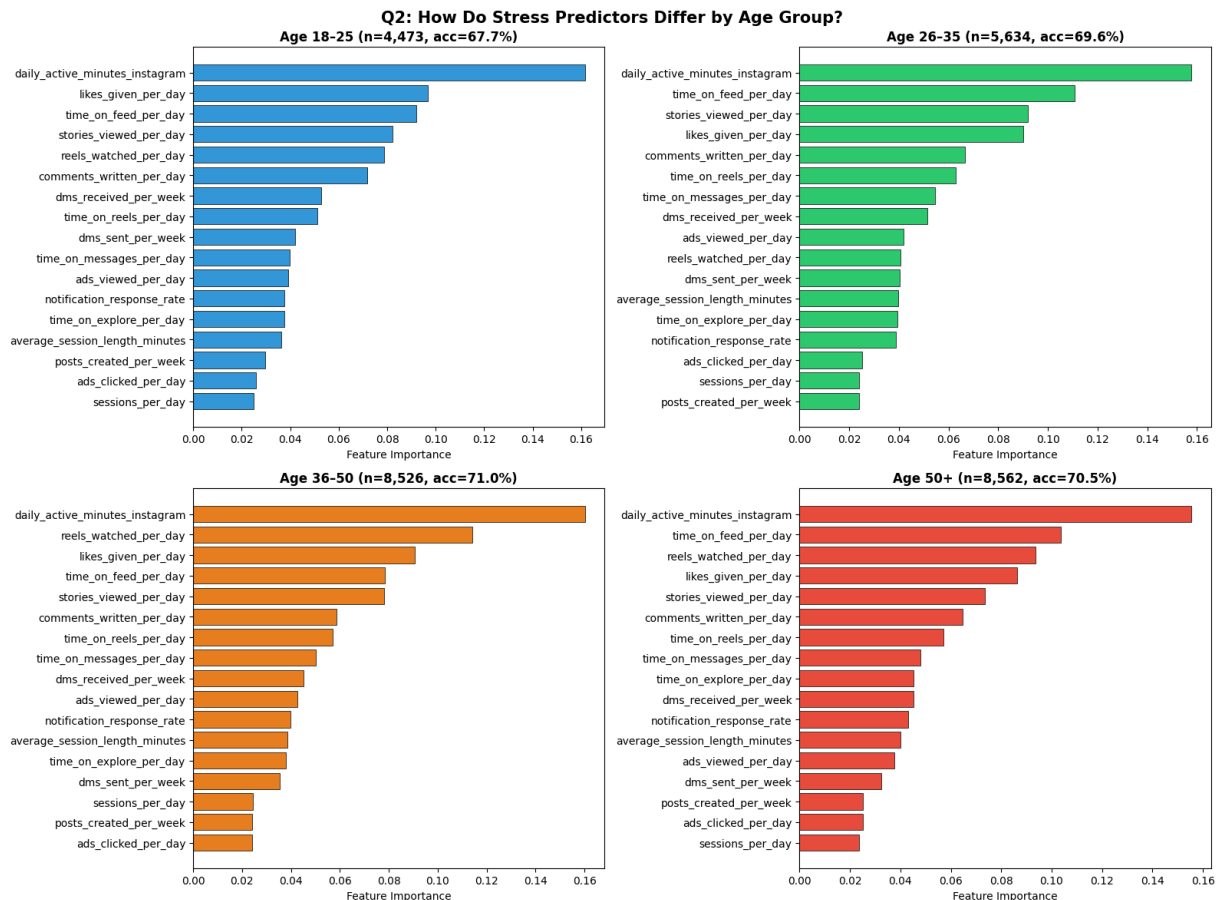
```

imp = data['importances'].sort_values(ascending=True)

ax.barh(imp.index, imp.values, color=age_colors[idx], edgecolor='black', linewidth=1)
ax.set_title(f"Age {label} (n={data['n']:,}, acc={data['accuracy']:.1%})",
            fontsize=12, fontweight='bold')
ax.set_xlabel('Feature Importance', fontsize=10)

plt.suptitle('Q2: How Do Stress Predictors Differ by Age Group?',
            fontsize=15, fontweight='bold')
plt.tight_layout()
plt.show()

```



Model 5: K Means Cluster Visualization

The left panel shows four behavioral segments spread cleanly along PC1 which captures 70% of the variance and represents overall usage intensity. The right panel colors the exact same projection by actual perceived stress score. The similarity between the two is clear: green (low stress) sits on the left and transitions smoothly to red (high stress) on the right. This confirms that clusters discovered purely from Instagram behavior with zero stress data as input, map directly onto mental health outcomes. Usage patterns alone are enough to approximate a user's stress level.

```

In [ ]: fig, axes = plt.subplots(1, 2, figsize=(16, 6))

cluster_colors = {0: '#3498db', 1: '#2ecc71', 2: '#e67e22', 3: '#e74c3c'}

```

```

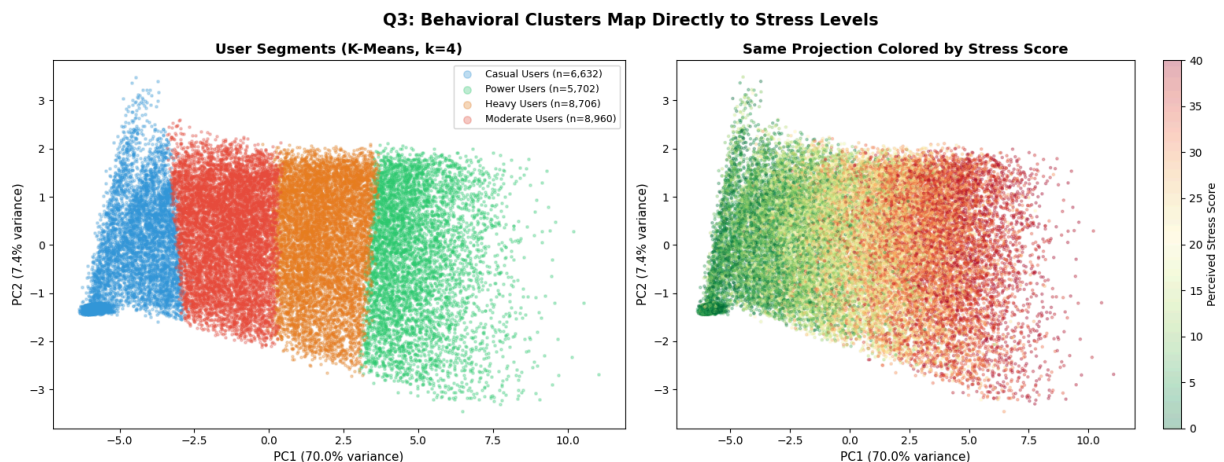
for c in sorted(df['cluster'].unique()):
    mask = df['cluster'] == c
    axes[0].scatter(df.loc[mask, 'pca_1'], df.loc[mask, 'pca_2'],
                    c=cluster_colors[c], alpha=0.3, s=5,
                    label=f'{cluster_names[c]} (n={mask.sum():,})')

axes[0].set_xlabel(f'PC1 ({pca.explained_variance_ratio_[0]:.1%} variance)', fontsize=13)
axes[0].set_ylabel(f'PC2 ({pca.explained_variance_ratio_[1]:.1%} variance)', fontsize=13)
axes[0].set_title('User Segments (K-Means, k=4)', fontsize=13, fontweight='bold')
axes[0].legend(fontsize=9, markerscale=3)

scatter = axes[1].scatter(df['pca_1'], df['pca_2'],
                          c=df['perceived_stress_score'],
                          cmap='RdYlGn_r', alpha=0.3, s=5)
axes[1].set_xlabel(f'PC1 ({pca.explained_variance_ratio_[0]:.1%} variance)', fontsize=13)
axes[1].set_ylabel(f'PC2 ({pca.explained_variance_ratio_[1]:.1%} variance)', fontsize=13)
axes[1].set_title('Same Projection Colored by Stress Score', fontsize=13, fontweight='bold')
cbar = plt.colorbar(scatter, ax=axes[1])
cbar.set_label('Perceived Stress Score', fontsize=10)

plt.suptitle('Q3: Behavioral Clusters Map Directly to Stress Levels',
              fontsize=15, fontweight='bold')
plt.tight_layout()
plt.show()

```



Model 6: Cluster Profiles (Q3)

The four segments form a clear gradient. Casual Users average just 42 minutes a day on Instagram and report the lowest stress (6.8/35) and highest happiness (7.0/10). At the other end, Power Users spend 346 minutes a day on the app, nearly 6 hours, and their stress is almost 5x higher (33.4/35) while their happiness drops to 3.8/10. The middle two groups follow the pattern predictably, but the jump from Heavy to Power Users is especially notable: daily usage only increases by about 100 minutes, but happiness drops by 1.5 points and stress jumps by 7 points. That suggests there may be a tipping point around 4+ hours of daily usage where the mental health effects accelerate. None of this was used as input to the clustering algorithm. The model grouped users purely by their behavioral patterns, and the stress and happiness differences emerged on their own.

```

In [ ]: fig, axes = plt.subplots(1, 3, figsize=(18, 5))
cluster_order = profiles_df['cluster'].astype(int).tolist()
ordered_names = [cluster_names[c] for c in cluster_order]
x = np.arange(len(ordered_names))
bar_colors = ['#2ecc71', '#3498db', '#e67e22', '#e74c3c']

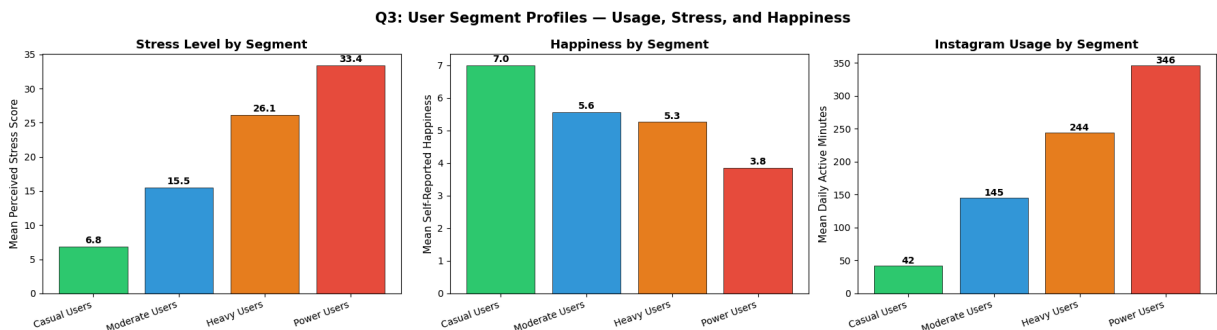
axes[0].bar(x, profiles_df['stress'].values, color=bar_colors, edgecolor='black', 1
axes[0].set_xticks(x)
axes[0].set_xticklabels(ordered_names, rotation=20, ha='right')
axes[0].set_ylabel('Mean Perceived Stress Score', fontsize=11)
axes[0].set_title('Stress Level by Segment', fontsize=13, fontweight='bold')
for i, v in enumerate(profiles_df['stress'].values):
    axes[0].text(i, v + 0.5, f'{v:.1f}', ha='center', fontweight='bold')

axes[1].bar(x, profiles_df['happiness'].values, color=bar_colors, edgecolor='black'
axes[1].set_xticks(x)
axes[1].set_xticklabels(ordered_names, rotation=20, ha='right')
axes[1].set_ylabel('Mean Self-Reported Happiness', fontsize=11)
axes[1].set_title('Happiness by Segment', fontsize=13, fontweight='bold')
for i, v in enumerate(profiles_df['happiness'].values):
    axes[1].text(i, v + 0.1, f'{v:.1f}', ha='center', fontweight='bold')

axes[2].bar(x, profiles_df['daily_min'].values, color=bar_colors, edgecolor='black'
axes[2].set_xticks(x)
axes[2].set_xticklabels(ordered_names, rotation=20, ha='right')
axes[2].set_ylabel('Mean Daily Active Minutes', fontsize=11)
axes[2].set_title('Instagram Usage by Segment', fontsize=13, fontweight='bold')
for i, v in enumerate(profiles_df['daily_min'].values):
    axes[2].text(i, v + 3, f'{v:.0f}', ha='center', fontweight='bold')

plt.suptitle('Q3: User Segment Profiles – Usage, Stress, and Happiness',
              fontsize=15, fontweight='bold')
plt.tight_layout()
plt.show()

```



Part 7: Insights and Conclusions

This project set out to explore whether Instagram usage patterns have a meaningful connection to mental health, specifically perceived stress. We approached this through three research questions, using both supervised and unsupervised machine learning techniques on a dataset of 90,000 Instagram users.

What we found:

Our classification models achieved around 70% accuracy when predicting whether a user falls into a Low, Medium, or High stress category based solely on their Instagram behavior. That's roughly double the 34.5% baseline, which tells us there is a real, measurable link between how someone uses Instagram and how stressed they are. The strongest predictor by far was daily active minutes, followed by time spent on the feed, likes given, and stories viewed. A pattern that stood out across every analysis we ran: passive consumption behaviors (scrolling, viewing, watching) are more predictive of stress than active behaviors (posting, commenting, creating). This aligns with existing psychology research suggesting that passive social media use is more harmful than active participation.

When we broke the analysis down by age group, the overall accuracy stayed consistent (67 to 71% across all groups), but the secondary predictors shifted. Younger users showed stronger connections between social validation features like likes and stress, while older users showed stronger connections between passive video consumption (reels) and stress. This suggests that while Instagram affects everyone, it may affect different age groups through different mechanisms.

Another finding came from our unsupervised analysis. K-Means clustering grouped users into four behavioral segments without any knowledge of their stress or happiness levels. When we checked those segments' mental health metrics after the fact, the results were striking: Casual Users (42 min/day) reported stress of 6.8/35 and happiness of 7.0/10, while Power Users which are heavy instagram users (346 min/day) reported stress of 33.4/40 and happiness of 3.8/10. The fact that behavioral grouping alone produces clusters with such dramatically different mental health profiles is strong evidence that the relationship between usage and wellbeing is not coincidental.

Limitations:

Our analysis used a sample of 30,000 users out of a dataset containing over 1 million entries. While 30,000 is more than sufficient for training reliable models, it's possible that sampling introduced slight biases that wouldn't exist if we ran the full dataset. We chose this sample size due to GitHub's file size limitations for hosting the final tutorial, but a production analysis with access to larger storage should use the complete data.

What this means:

App designers could use behavioral signals to identify users who may be at risk of negative mental health outcomes and create interventions like usage reminders or content breaks.

Researchers studying social media and wellbeing could use classification models like ours to move beyond simple screen time metrics and look at specific behavioral patterns. And for individual users, the finding that passive scrolling is more strongly linked to stress than active creation suggests a practical takeaway: if you're going to spend time on Instagram, creating and engaging may be healthier than endlessly scrolling, otherwise your happiness and stress levels will worsen.

How readers will interpret our findings

We believe that readers who previously were uninformed about the topic will now feel more informed about the topic once he/she reads through the project because they would see us examine usage patterns of Instagram users and connections to lifestyle factors and wellbeing metrics, therefore informing them about how certain Instagram behaviors, such as spending more time on Instagram every day, tend to bring out more stress in their users as well as making them less happy. People who are already informed will probably still learn something new, as while they may already have the belief that spending more time on Instagram will reduce users' general happiness, they will still learn more specific behavioral patterns for demographics like age that also factor into our findings, such as finding that people from 36-50 years old tend to consume more reels per day than those from 18-25 years old.